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A PROJECT BASED LEARNING

CASE STUDY REPORT ON

Student Performance Analytics

Submitted by

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For the course: Data Warehousing and Mining

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B. Tech Computer Engineering (Sem V)

2026–27



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CERTIFICATE

This is to certify that the project report entitled "Data Deli Manager Performance Dashboard – Case Study" is a bonafide work carried out by Satyam Kumar Singh (87) in partial fulfilment of the requirements for the award of the degree of B. Tech in Computer Engineering during the academic year 2026–27. The work has been completed under my guidance and supervision. To the best of my knowledge, the work presented in this report has not been submitted, either in whole or in part, to any other university or institution for the award of any degree or diploma.

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ACKNOWLEDGEMENT

We express our sincere gratitude to Dr. Rajesh Prasad, Principal of Bharati Vidyapeeth (Deemed to be University), College of Engineering, Pune, for providing the facilities and academic environment required for completing this project.

We are deeply thankful to Dr. Sandeep Vanjale, Head of the Department of Computer Engineering, for valuable guidance and encouragement throughout the project work. We extend our heartfelt gratitude to our project guide, Dr. Rohini Khalkar, for continuous guidance, constructive suggestions, technical support, and motivation during every stage of the project.

We also thank all faculty members, friends, and family members who supported us during the development and documentation of the Data Deli Manager Performance Dashboard. Finally, we thank our institution for providing an opportunity to undertake this project and apply concepts of data warehousing, data mining, business intelligence, and data visualization in a practical retail-management application.

Ayush Kumar

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ABSTRACT

Retail and food-service managers require timely and reliable information to track sales, revenue, stock levels, product performance, supplier behaviour and unusual events. When this information is scattered across spreadsheets, point-of-sale exports and separate systems, managers spend more time preparing reports than acting on them. The **Data Deli Manager Performance Dashboard**, deployed as **Meridian WMS – Grocery Warehouse Data Mining**, is a web-based business-intelligence and data-mining prototype built to address this problem.

The system accepts a CSV or Excel file of grocery and deli products and automatically applies five data-mining techniques: **k-Nearest-Neighbour (kNN) classification**, **k-means clustering**, **Apriori association-rule mining**, **linear regression** and **z-score outlier detection**. On the bundled dataset of **3,133 products (SKUs) across 15 categories and 21 numeric attributes**, the dashboard assigns every product a predicted category, groups the products into four clusters, discovers **296 association rules** (average confidence 81.7%), fits a sales trend line (**Monthly_Sales = 0.515 × Quantity_In_Stock + 45.70, R^2 = 0.52**) and flags **52 outlier products (1.7%)** whose sales deviate strongly from the trend.

This case study documents the project end to end: requirement analysis, layered architecture, a conceptual data-warehouse (star-schema) design, algorithm descriptions, implementation, testing and a critical evaluation of the results. The evaluation is based on the figures reported by the deployed application and on values derived from them. It shows, for example, that kNN accuracy of 16.8% is about 2.5 times the random-guess level for 15 classes, that k-means clusters (silhouette 0.08) reflect stock-and-sales behaviour rather than product category, and that the highest-confidence association rules mostly restate relationships between derived attributes. The report closes with limitations and a roadmap for improvement.

The project deployment link is https://data-deli-manager-performance.vercel.app/.

Keywords: Business intelligence, deli management, performance dashboard, data warehousing, data mining, kNN, k-means, Apriori, linear regression, outlier detection, retail analytics, visualization.

1. INTRODUCTION

1.1 Background

Deli and grocery operations involve frequent transactions, short shelf-life products, supplier lead times, fluctuating demand and tight cost control. A change in one area, such as a delayed delivery or an over-ordered perishable item, quickly affects revenue, waste and customer satisfaction. Managers therefore need to see stock, sales, pricing and product behaviour together rather than in isolated reports.

A performance dashboard presents selected measures visually and reduces the effort needed to interpret raw records. Data mining goes one step further: instead of only summarising what happened, it searches the data for patterns such as groups of similar products, rules about what goes together, relationships between variables and records that behave abnormally. The Data Deli Manager Performance Dashboard combines both ideas in a single browser-based tool.

1.2 Problem Statement

Deli managers hold large product files containing prices, costs, stock levels, sales, discounts, expiry periods and supplier details, but these files are rarely analysed beyond simple totals. Slow-moving products, unusually strong sellers, mis-priced items and hidden relationships between attributes remain unnoticed until the end of a reporting period.

The central problem addressed by this project is: **How can a web-based dashboard take a raw product dataset and automatically present classification, clustering, association, trend and outlier insights that help managers make faster, evidence-based decisions?**

1.3 Objectives

- To develop a web-based dashboard for deli and grocery performance monitoring that works on any uploaded CSV / XLSX dataset.
- To organise product, stock, sales and supplier data into a structure suitable for analysis and historical reporting (data-warehouse view).
- To apply kNN classification, k-means clustering, Apriori association rules, linear regression and z-score outlier detection and present their results visually.
- To provide interactive controls (axes, k, minimum support, minimum confidence, z threshold) so that managers can explore the effect of parameters.
- To classify, cluster and outlier-check newly added products so that new stock is assessed immediately.
- To evaluate each technique with suitable metrics and to discuss critically what the results do and do not show.
- To demonstrate data-warehousing and data-mining concepts of the course (Data Warehousing and Mining) in a realistic retail setting.

1.4 Scope

The system is an analytical prototype. It focuses on exploring and mining product-level data and presenting decision-support information. It does not replace point-of-sale, accounting, payroll or full inventory-control systems.

- Dataset upload (CSV / XLSX), bundled sample dataset and a "new batch" file for new products.
- Dataset summary: number of SKUs, number of categories, attribute list and headline mining metrics.
- Five mining modules: classification, clustering, association rules, regression and outlier detection.
- Product ledger listing every product with its actual and predicted category, confidence, cluster, key measures and status, with filters and export.
- Conceptual star-schema design for historical, multi-dimensional analysis.

1.5 Need for the System

Retail managers must react quickly to changing demand, product availability, pricing and supplier performance. Manual spreadsheet analysis is slow, error-prone and rarely reveals non-obvious structure. A dashboard that automatically mines the data lets managers focus on action – promoting a slow mover, investigating a product that sells far above its stock level, or revising reorder points – instead of on report preparation.

1.6 Project at a Glance

Table 1: Project summary

Item	Details
Project title	Data Deli Manager Performance Dashboard
Deployed application name	Meridian WMS – Grocery Warehouse Data Mining
Deployment	https://data-deli-manager-performance.vercel.app/ (Vercel)
Dataset	smart_grocery_warehouse_dataset.csv – 3,133 SKUs, 15 categories, 21 numeric attributes
Mining techniques	kNN, k-means, Apriori, linear regression, z-score outliers
Headline results	kNN accuracy 16.8% · 4 clusters (silhouette 0.08) · 296 rules · R^2 0.52 · 52 outliers (1.7%)
Course	Data Warehousing and Mining, B.Tech Computer Engineering (Sem V)

1.7 Organization of the Report

Chapter 1 introduces the project. Chapter 2 presents the literature survey. Chapter 3 discusses requirements, the dataset and feasibility. Chapter 4 explains system design, including architecture, data flow and the warehouse schema. Chapter 5 describes the methodology, algorithms and

implementation. Chapter 6 presents testing, results and a critical analysis with graphs. Chapter 7 concludes the case study and outlines limitations and future scope.

2. LITERATURE SURVEY

2.1 Introduction

Business-intelligence dashboards give managers visual access to key information. In retail and food-service they typically combine point-of-sale data, inventory indicators, customer data and financial metrics. The concepts relevant to this project are retail performance measurement, data warehousing, data mining (classification, clustering, association, regression and outlier detection), KPI design and visual analytics.

2.2 Retail Performance Dashboards

A retail-performance dashboard presents metrics that explain commercial and operational health: sales, number of orders, average order value, sales by category, product contribution, discounts and margin. Few (2006) argues that a dashboard succeeds when it supports at-a-glance monitoring, highlighting changes and exceptions rather than merely listing totals. This principle guides the design of the overview tiles and the outlier panel in this project.

2.3 Data Warehousing in Retail

A data warehouse integrates historical data from operational sources into a structure optimised for analysis. Kimball and Ross recommend dimensional modelling: a central fact table of numeric measures surrounded by descriptive dimension tables such as date, product, category, brand and supplier. This star schema supports fast aggregation and consistent drill-down. In this project the schema is used as a conceptual model for the product-level data and as the basis for future historical analysis.

2.4 Data Mining Techniques Used

Classification. Classification assigns a record to a predefined class. The k-nearest-neighbour rule of Cover and Hart (1967) predicts the class that is most common among the k most similar training records; it needs no training phase but depends strongly on the quality of the distance measure and the separability of classes.

Clustering. Clustering groups records without predefined labels. The k-means algorithm (MacQueen, 1967; Lloyd, 1982) repeatedly assigns records to the nearest of k centroids and recomputes the centroids. The silhouette coefficient of Rousseeuw (1987) measures how well each record fits its cluster compared with the nearest other cluster.

Association rules. Agrawal and Srikant (1994) introduced the Apriori algorithm, which finds item combinations that occur frequently and derives rules of the form $A \rightarrow B$ measured by support, confidence and lift. Lift greater than 1 indicates that A and B occur together more often than if they were independent.

Regression and outliers. Linear regression models a numeric target as a linear function of one or more predictors and is evaluated with R^2. Residual analysis then identifies records that deviate strongly from the fitted trend; a standardised residual (z-score) above a chosen threshold, commonly 2.5 to 3, marks a potential outlier (Hawkins, 1980).

2.5 KPI-Based Decision Support

Key performance indicators translate business goals into measurable quantities. A good KPI is relevant, reliable, understandable, timely and linked to an action. For example, a product with high stock but low sales may require a pricing or procurement decision, while a product selling far above its stock-based expectation may need an earlier reorder.

Table 2: Comparative analysis of retail analytics techniques

Technique	Data input	Managerial use	Strength	Limitation
Descriptive analytics	Sales and order records	Summarise past performance	Easy to interpret	Does not predict
Classification (kNN)	Labelled product attributes	Auto-categorise new products	Simple, no training	Weak if classes overlap
Clustering (k-means)	Numeric attributes	Segment products by behaviour	Finds hidden groups	Needs k; results need interpretation
Association rules (Apriori)	Discretised attributes / baskets	Find co-occurring conditions	Readable if-then rules	Many trivial rules
Regression	Numeric predictor and target	Explain and forecast sales	Transparent equation	Assumes linear relation
Outlier detection (z-score)	Regression residuals	Spot unusual products	Fast and explainable	Sensitive to threshold and distribution

2.6 Research Gap

Basic sales applications record transactions, while many dashboards show only high-level totals. Academic data-mining tools, in contrast, are usually desktop packages that require expert knowledge. A manager-oriented solution should bring mining results – categories, groups, rules, trends and exceptions – into one browser-based workspace that works on the manager's own file. This project addresses the gap with a single page that mines an uploaded product dataset with five techniques and keeps the results connected through a product ledger.

3. REQUIREMENT ANALYSIS AND FEASIBILITY STUDY

3.1 Existing System

In a conventional deli or grocery environment, product master data and sales are exported from a point-of-sale or stock system and reviewed in spreadsheets. Supplier details, expiry information and customer feedback often live in separate files. This creates several limitations: report preparation takes time, categories are applied inconsistently, comparisons are delayed, and patterns such as unusual sellers or redundant relationships are not noticed.

3.2 Proposed System

The proposed dashboard provides a centralised interface that ingests a product dataset and immediately produces dataset statistics, mining results and a product ledger. The manager can change the parameters of each technique, inspect the resulting charts and rule lists, and add a new batch of products to see how they are classified, clustered and checked for outliers.

3.3 Functional Requirements

Table 3: Functional requirements

ID	Functional requirement
FR-01	The system shall load in a web browser without installation.
FR-02	The system shall accept a CSV or Excel dataset with at least two numeric columns and ideally a category column.
FR-03	The system shall provide a sample dataset (sample.csv) and a sample new batch (new_batch.csv).
FR-04	The system shall display the number of SKUs, categories and attributes, and headline metrics for each technique.
FR-05	The system shall classify products into categories using kNN (k = 5) and report accuracy and F1-score.
FR-06	The system shall cluster products using k-means with a user-selectable k and selectable X/Y axes.
FR-07	The system shall mine association rules with adjustable minimum support and minimum confidence, showing support, confidence and lift.
FR-08	The system shall fit a linear-regression trend, show the equation and R^2, and flag outliers above a user-selectable $\|z\|$ threshold.
FR-09	The system shall list all products in a ledger with actual and predicted category, confidence, cluster, key measures and status, filterable by All / New / Outliers.
FR-10	The system shall allow new products to be added and automatically assigned a class, a cluster and an outlier status.
FR-11	The system shall allow the ledger to be exported.
FR-12	The system shall handle missing or invalid data without failure and show understandable messages.

3.4 Non-Functional Requirements

Table 4: Non-functional requirements

ID	Non-functional requirement
NFR-01	The interface should be simple and understandable for managers without data-science training.

ID	Non-functional requirement
NFR-02	Mining results for a dataset of a few thousand rows should appear with acceptable response time.
NFR-03	Numbers shown in tiles, charts, rule lists and the ledger must be consistent with each other.
NFR-04	The system should be modular so that techniques can be added or replaced.
NFR-05	Uploaded business data should be protected and not shared; production use needs authentication.
NFR-06	The layout should be responsive on desktop and mobile screens.
NFR-07	The design should allow more records, attributes and techniques.
NFR-08	Charts should be readable with sufficient contrast and clear labels.

3.5 Dataset Description

The bundled dataset **smart_grocery_warehouse_dataset.csv** describes **3,133 products across 15 categories**, such as vegetables, pulses, cooking oil, spices, dairy, bakery, packaged foods, beverages, household and cleaning products, snacks and personal care. Product names carry a brand prefix (FreshMart, ValueCart, DailyChoice, PrimePick, NatureNest, FarmFresh). Each product has **21 numeric attributes**, listed below with their meaning (definitions are inferred from the attribute names).

Table 5: Attribute dictionary of the dataset

Attribute	Group	Meaning
Unit_Price	Pricing	Purchase price per unit from the supplier
Cost_Price	Pricing	Cost to the store of one unit
Selling_Price	Pricing	Price charged to the customer
Discount_Percentage	Pricing	Average discount applied (%)
Profit_Margin	Profitability	Share of selling price retained as profit
Quantity_In_Stock	Inventory	Units currently in the warehouse
Minimum_Stock_Level	Inventory	Reorder point
Maximum_Stock_Level	Inventory	Upper stock limit
Restock_Frequency_Days	Inventory	Days between restocking
Supplier_Lead_Time_Days	Supplier	Days a supplier takes to deliver
Units_Sold	Sales	Units sold in the period
Monthly_Sales	Sales	Sales volume per month (target of regression)
Sales_Velocity	Sales	Rate at which stock sells (derived from sales)
Revenue	Sales	Income generated by the product
Profit	Profitability	Revenue minus cost
Basket_Quantity	Customer behaviour	Typical quantity per customer basket
Purchase_Frequency	Customer behaviour	How often customers buy the product
Customer_Rating	Customer behaviour	Average customer rating
Expiry_Days	Product	Shelf life in days
Storage_Temperature_C	Product	Required storage temperature (°C)
Product_Weight_Kg	Product	Weight of one unit (kg)

In addition, the mining modules create derived, discretised attributes such as Monthly_Sales = Low / Mid / High, Sales_Velocity = Low / Mid / High and Demand_Level = Low / Mid / High, which appear in the association rules (Section 6.4.4).

3.6 Feasibility Study

Technical feasibility: Modern browsers can parse CSV/XLSX files, compute distances, centroids, regressions and rule support for datasets of a few thousand rows, and draw interactive charts. Cloud platforms such as Vercel host the application with no server administration.

Economic feasibility: The prototype is built with open-source tools and free hosting, so development and deployment costs are negligible for academic use.

Operational feasibility: Managers are familiar with spreadsheets, tables and charts. Uploading a file and reading tiles and rankings fits their routine; parameter sliders keep technical choices optional.

Schedule feasibility: The work can be done in phases – requirements, data preparation, interface, algorithms, testing, deployment and documentation – within one semester.

3.7 Use-Case Description

The primary actor is the **deli / warehouse manager**. The manager opens the dashboard, uploads a dataset or loads the sample, reads the dataset summary and headline metrics, explores the clusters and regression chart by changing axes and parameters, reviews association rules and outliers, and finally filters the product ledger. Secondary actors include an **administrator** who prepares and maintains data files, an **analyst** who interprets the mining results, and a **staff member** who may use a read-only view.

Table 6: Main use cases

Use case	Actor	Outcome
Upload or load dataset	Manager / Admin	Dataset summary and mining metrics appear
Explore clusters and regression	Manager / Analyst	Scatter plots update for chosen axes and k
Review association rules	Analyst	Rules sorted with support, confidence and lift
Investigate outliers	Manager	Products unusually far from the sales trend are listed
Add new products	Manager	New items are classified, clustered and checked
Filter and export ledger	Manager / Analyst	All / New / Outlier product lists can be exported

4. SYSTEM DESIGN

4.1 System Architecture

The dashboard uses a layered architecture. The **presentation layer** contains the upload bar, navigation (Overview, Performance, Rules, Regression, Outliers, Products), parameter sliders, tiles, charts and the product ledger. The **application layer** parses uploaded files, validates them, loads the sample data and keeps the shared parameter state. The **analytics layer** runs the five data-mining techniques and the evaluation metrics. The **data layer** holds the dataset file, the sample file and the new-batch file.

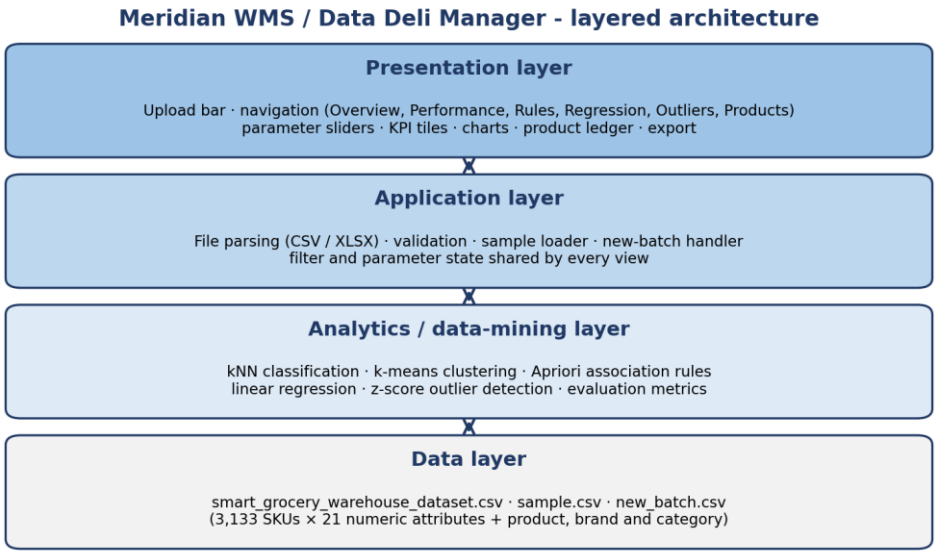


Figure 4.1: Layered system architecture of the dashboard

4.2 Data-Flow Description

A dataset enters through file upload or the “Load sample” button. The application checks that at least two numeric columns exist, prepares the records and passes them to the five mining modules. The results – metrics, cluster labels, rules, trend line and outlier flags – are stored once and read by every view, which keeps tiles, charts and the ledger consistent. When a new batch is added, each new product is scored with the already prepared models.

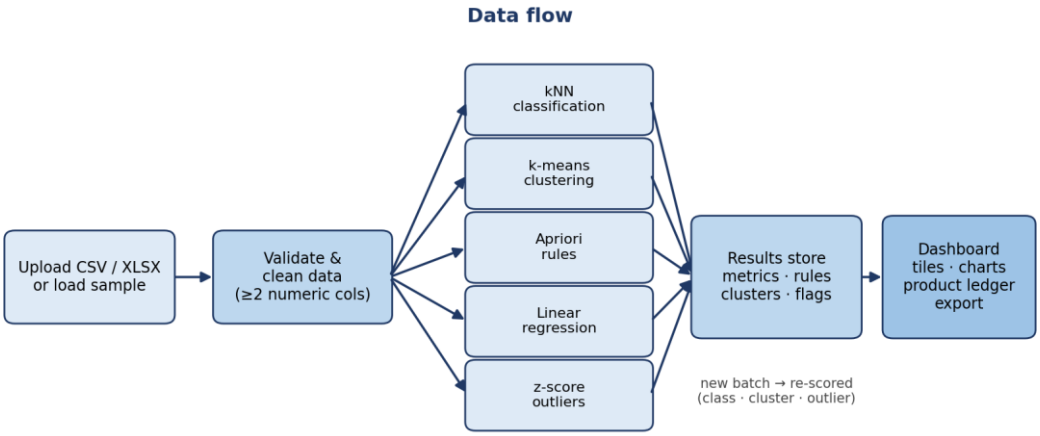


Figure 4.2: Data-flow diagram

1. Upload a CSV / XLSX file or load the sample dataset.
2. Validate the file (numeric columns, readable rows) and clean obvious problems.
3. Prepare numeric attributes and, where needed, discretise them into Low / Mid / High.
4. Run kNN, k-means, Apriori, regression and outlier detection.
5. Store metrics, labels, rules and flags in one shared results structure.
6. Display tiles, charts, rule lists, outlier list and the product ledger.
7. Optionally add a new batch of products and re-score only the new rows.

4.3 Data Warehouse Design

A star schema is suitable for performance analysis because it organises measurable events around descriptive dimensions. The dashboard itself works on a flat product file, but the same attributes map naturally to the conceptual warehouse shown below, which would allow historical, multi-period analysis when transaction data is available.

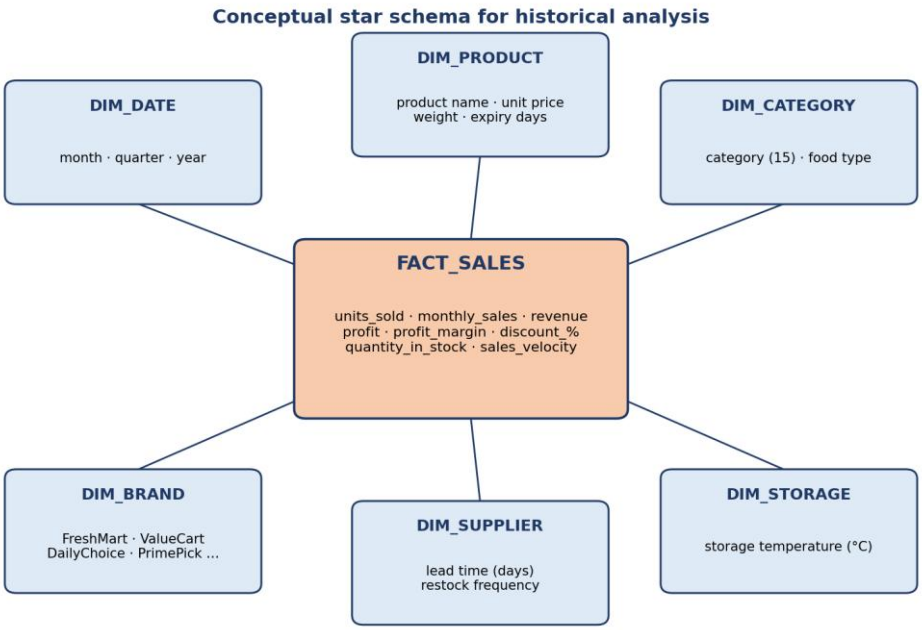


Figure 4.3: Conceptual star schema for product performance

Table 7: Sales fact table (conceptual)

Fact / attribute	Description
sales_id	Unique identifier of the fact row
date_id, product_id, category_id, brand_id, supplier_id, storage_id	Foreign keys to the dimension tables
units_sold	Units sold in the period
monthly_sales	Sales volume per month
revenue / profit / profit_margin	Financial measures
discount_percentage	Discount applied
quantity_in_stock	Stock level at the time of measurement
sales_velocity	Rate of sale (derived measure)

Table 8: Dimension tables (conceptual)

Dimension	Example attributes
Date	Date, month, quarter, year
Product	Product name, unit price, weight, expiry days
Category	Category name (15 categories), food type
Brand	FreshMart, ValueCart, DailyChoice, PrimePick, NatureNest, FarmFresh
Supplier	Supplier lead time, restock frequency
Storage	Storage temperature (°C)

4.4 User-Interface Design

The interface is a single scrolling page with a dark theme and a sticky top bar. The top bar shows the product name “MERIDIAN WMS”, section links and a “warehouse · live” status. Below it, an

upload bar accepts a dataset by drag-and-drop or click, with buttons for "+ Add new products" and "Load sample" and the example files sample.csv and new_batch.csv. A dataset summary states the number of SKUs and categories, lists the attributes, and shows four headline tiles: clusters, accuracy, rules and outliers.

- **Overview:** dataset summary, headline tiles and the "Data Mining Performance" bar chart on a common 0–100 scale.
- **Performance:** one metric card per technique (kNN, k-means, Apriori, regression, outliers).
- **Clustering and regression panels:** axis selectors and sliders for k, minimum support, minimum confidence and |z| threshold.
- **Rules and outliers panels:** scrollable lists of association rules and flagged products.
- **Product ledger:** table with All / New / Outliers filters and an Export button.

4.5 KPI and Analytics Design

The dashboard reports technique-specific metrics and derives business measures from the dataset. The main formulas are listed below.

$$\text{Profit Margin (\%)} = (\text{Profit} / \text{Revenue}) \times 100$$

$$\text{Euclidean distance } d(x, y) = \sqrt{\Sigma (x_i - y_i)^2}$$

$$\text{Support}(A \rightarrow B) = P(A \cap B)$$

$$\text{Confidence}(A \rightarrow B) = P(A \cap B) / P(A) \qquad \text{Lift} = \text{Confidence} / P(B)$$

$$\text{Regression: } \hat{y} = b_1 x + b_0 \qquad \text{Residual } e = y - \hat{y} \qquad z = e / \sigma_e$$

$$R^2 = 1 - SS_{res} / SS_{tot} \qquad \text{Silhouette } s = (b - a) / \max(a, b)$$

$$\text{Accuracy} = \text{correct predictions} / \text{all predictions}$$

$$F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$$

5. METHODOLOGY AND IMPLEMENTATION

5.1 Development Methodology

The project follows an incremental methodology. Each mining module was developed and checked as a separate unit before being integrated into the dashboard, which made it possible to verify calculations, visual components and data flow step by step.

1. Requirement gathering and definition of performance measures.
2. Study of dashboard patterns and retail data-mining techniques.
3. Design of the data model and analytical dimensions.
4. Dataset collection and preparation.
5. Development of interface components (upload bar, tiles, charts, ledger).
6. Implementation of the five mining algorithms and their metrics.
7. Integration of charts, rule lists, outlier list and ledger.
8. Testing, responsive checks and parameter experiments.
9. Deployment on Vercel and documentation.

5.2 Data Collection

The analysis uses a grocery warehouse dataset named **smart_grocery_warehouse_dataset.csv**. It contains 3,133 products, 15 categories and 21 numeric attributes (Table 5). The application also ships with sample.csv for quick demonstration and new_batch.csv to demonstrate how newly arriving products are handled. For an academic prototype the limitations of the data should be stated clearly: the dataset is a prepared sample, not a live feed from a store.

5.3 Data Preprocessing

- Validate that the file has at least two numeric columns; a category column is optional but enables classification.
- Convert text-formatted numbers to numeric values and ignore unreadable cells.
- Handle missing values so that distance and regression calculations do not fail.
- Remove duplicate or invalid records where detected.
- Standardise category names and product labels.
- Discretise Monthly_Sales, Sales_Velocity and Demand_Level into Low / Mid / High so that Apriori can treat them as items.
- Compute derived values such as regression residuals and z-scores for outlier detection.

5.4 Analytics Workflow

After the dataset is prepared, the analytics layer computes (i) a classification of every product, (ii) cluster assignments for the chosen k, (iii) association rules for the chosen support and confidence

thresholds, (iv) a regression line for the chosen axes, and (v) residual z-scores, from which outliers are flagged. Because all views read the same results, changing one parameter updates the corresponding panel and the ledger consistently.

5.5 Algorithm Description

5.5.1 kNN classification (target: Category)

For each product the algorithm finds the k = 5 nearest products by Euclidean distance over the numeric attributes and predicts the most frequent category among them. The dashboard evaluates the classifier with **leave-one-out validation**: every product is classified using all other products as training data, which gives an almost unbiased accuracy estimate and does not need a separate test set. Reported metrics are accuracy and F1-score over 15 classes. For new products the predicted category, a confidence value and the nearest cluster are assigned automatically.

1. Input: dataset X with numeric attributes, labels y, k = 5.
2. For each product i, compute distances to all other products $j \neq i$.
3. Select the 5 smallest distances and take a majority vote of their categories.
4. Confidence = share of the 5 neighbours that agree with the vote.
5. Compare the predicted and actual category; accumulate accuracy and F1.

5.5.2 k-means clustering

k-means partitions the products into k groups (default k = 4). It initialises k centroids, assigns each product to the nearest centroid, recomputes each centroid as the mean of its members, and repeats until assignments stop changing. The scatter plot shows clusters on two user-chosen axes (default Quantity_In_Stock × Monthly_Sales). Cluster quality is summarised by the **silhouette coefficient**, which ranges from −1 to +1; the dashboard rescales it to 0–100 as $(s + 1)/2 \times 100$ for the common performance chart.

5.5.3 Apriori association rules

Apriori mines rules of the form A → B among the discretised attributes. Only itemsets whose support is at least the **minimum support (default 0.10)** are kept, and only rules whose confidence is at least the **minimum confidence (default 0.60)** are reported. Each rule is shown with support, confidence and lift. The default settings produce **296 rules**.

5.5.4 Linear regression

Ordinary least-squares regression fits Monthly_Sales as a linear function of Quantity_In_Stock (axes can be changed). The fitted model reported by the dashboard is:

$$Monthly_Sales = 0.515 \times Quantity_In_Stock + 45.70 \qquad (R^2 = 0.52)$$

5.5.5 Outlier detection

Outliers are detected from the regression residuals. For every product the residual e = actual − predicted is divided by the standard deviation of all residuals to obtain a z-score; products with |z|

above the threshold (default 2.5) are flagged. The outlier panel lists them as “above trend” or “below trend” with actual and expected monthly sales.

Table 9: Summary of algorithms and parameters

Technique	Task	Key parameters	Metric reported	Result (dashboard)
kNN	Classify category	k = 5, leave-one-out	Accuracy, F1	16.8%, F1 14.7%, 15 classes
k-means	Cluster products	k = 4, axes selectable	Silhouette	0.08 (54.0 on 0–100)
Apriori	Association rules	min support 0.10, min confidence 0.60	Rules, avg confidence	296 rules, 81.7%
Linear regression	Explain sales	x = stock, y = monthly sales	R^2	0.52 (51.9%)
z-score outliers	Flag anomalies	$\|z\| > 2.5$	Records flagged	52 of 3,133 (1.7%)

5.6 Modules of the System

Table 10: Modules

Module	Responsibility
Data Management	Upload, parse, validate and prepare CSV / XLSX data; load sample and new-batch files
Overview / Dashboard	Dataset summary, headline tiles and the performance comparison chart
Classification (kNN)	Predict product category, confidence and nearest cluster
Clustering (k-means)	Group products and display the cluster scatter plot
Association Rules (Apriori)	Generate and list rules with support, confidence and lift
Regression	Fit and display the sales trend and equation
Outlier Detection	Compute z-scores and list flagged products
Product Ledger and Reporting	Tabular view with filters (All / New / Outliers) and export

5.7 Technology Stack

Table 11: Technology stack

Layer	Technology / implementation	Purpose
Frontend	Browser-based single-page web application	Dashboard interface and interactions
Styling	CSS with a dark responsive theme	Layout and visual design
Visualization	Interactive bar, scatter and table components	Charts, tiles and ledger
Data processing	Parsing of CSV / XLSX and in-application mining logic	kNN, k-means, Apriori, regression, z-score
Data storage	Uploaded file and bundled sample files	Input data and new-batch data
Deployment	Vercel	Public web hosting
Version control	Git and GitHub	Source-code management

5.8 Feature List

Table 12: Feature list

Feature	Description
Drag-and-drop upload	Accepts CSV or Excel files with at least two numeric columns
Sample and new-batch files	One-click demonstration data
Dataset summary	SKU count, category count, attribute list
Headline tiles	Clusters, accuracy, rules and outliers at a glance
Normalised performance chart	Compares all five techniques on a 0–100 scale
Interactive parameters	Axes, k, minimum support, minimum confidence, |z| threshold
Cluster and regression plots	Scatter plots with colour-coded clusters and highlighted outliers
Association-rule list	Support, confidence and lift for every rule
Outlier list	Products ranked by z-score with actual vs expected sales
Product ledger	Actual vs predicted category, confidence, cluster, measures and status; All / New / Outliers filter; export

5.9 Implementation Details

The interface is built from reusable components: a header and upload bar, summary tiles, chart panels, list panels and a ledger table. Parameter controls update a shared state so that every panel is recomputed for the same context. Figure 5.1 shows the overview of the deployed application, Figure 5.2 shows the mining panels, and Figure 5.3 shows the product ledger.

The deployed application is available at https://data-deli-manager-performance.vercel.app/.

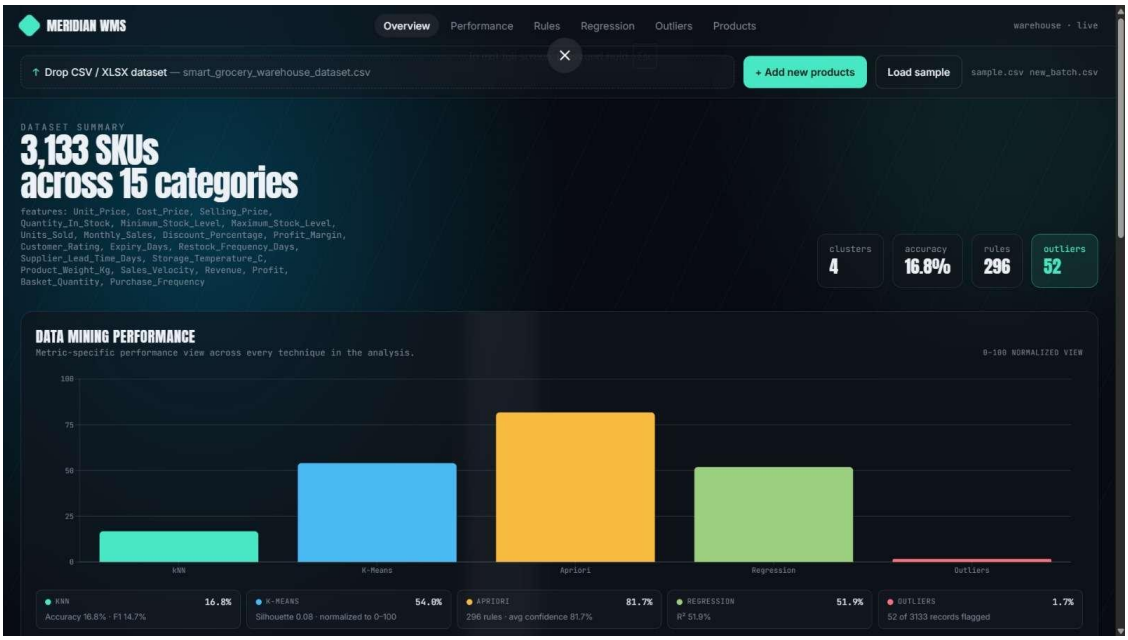


Figure 5.1: Overview – dataset summary, headline tiles and Data Mining Performance chart

Reading Figure 5.1. The dataset summary states 3,133 SKUs across 15 categories and lists all 21 attributes. The four tiles show 4 clusters, 16.8% classification accuracy, 296 rules and 52 outliers.

The bar chart normalises every technique to a 0–100 scale (kNN 16.8, k-means 54.0, Apriori 81.7, regression 51.9, outliers 1.7), and the cards beneath show the underlying metric of each technique.

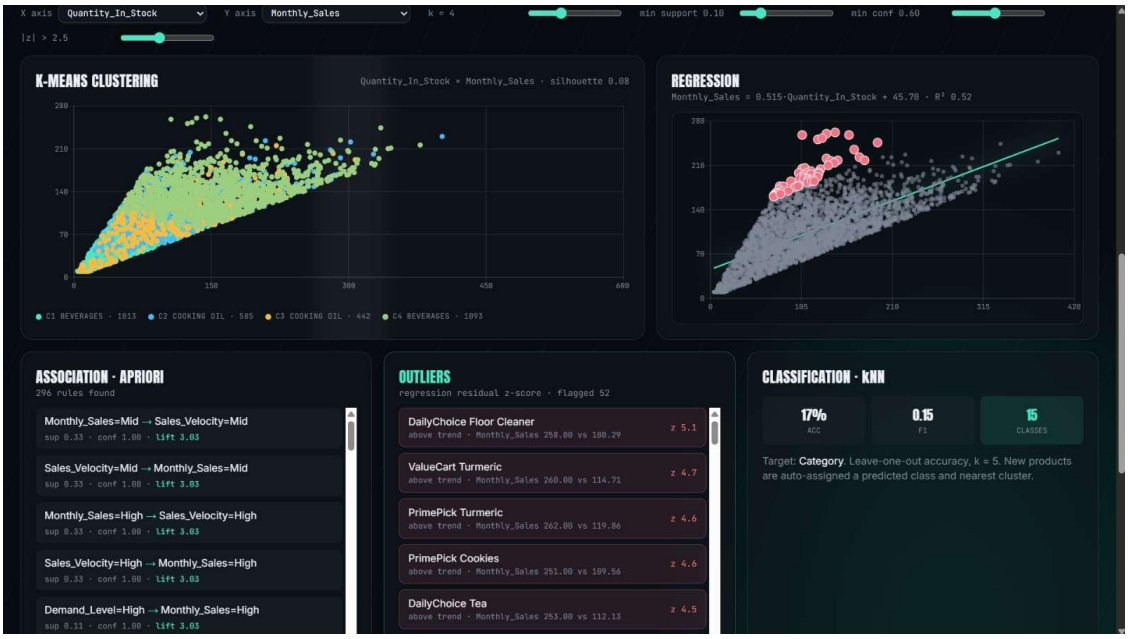


Figure 5.2: K-means clustering, regression, association rules, outliers and kNN panels

Reading Figure 5.2. The controls at the top set the axes (Quantity_In_Stock and Monthly_Sales), k = 4, minimum support 0.10, minimum confidence 0.60 and |z| > 2.5. The k-means panel lists the four clusters with their sizes (1,013, 585, 442 and 1,093 products). The regression panel shows the fitted equation and highlights flagged products in red. The lower panels show the association rules, the outlier list and the kNN summary.

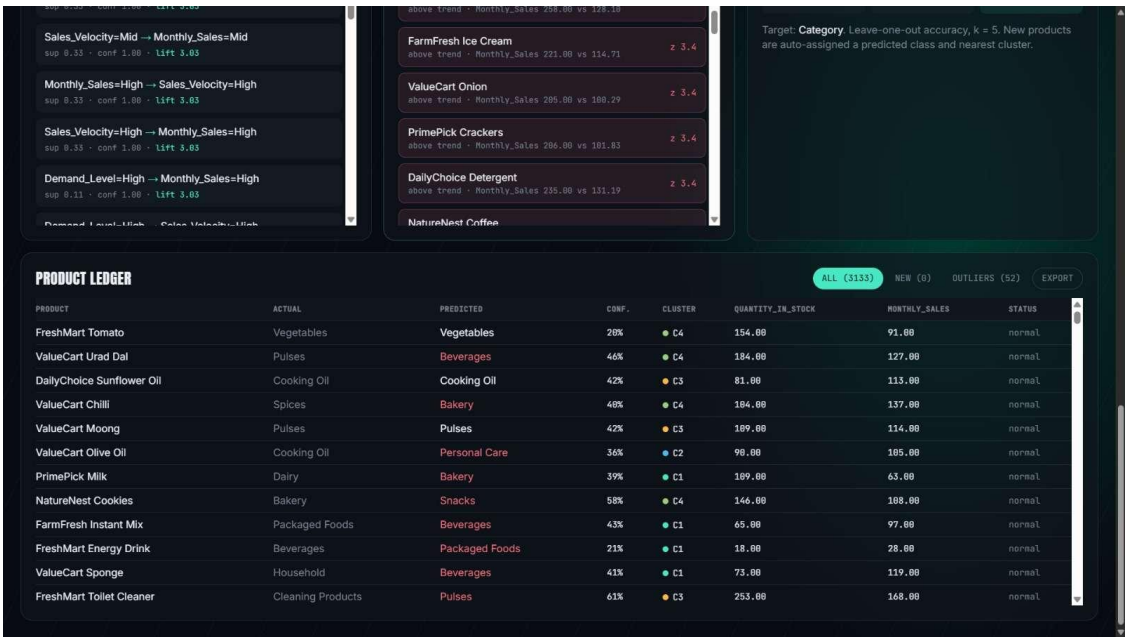


Figure 5.3: Product ledger with actual and predicted categories, clusters and status

Reading Figure 5.3. The ledger lists every product with its actual and predicted category, prediction confidence, cluster, quantity in stock, monthly sales and status. The filter chips show

ALL (3,133), NEW (0) and OUTLIERS (52). Mis-predicted categories are shown in red, which makes classification errors easy to spot.

6. TESTING AND RESULTS

6.1 Testing Strategy

The dashboard is evaluated through functional testing, calculation verification, integration testing, user-interface testing, responsive testing and data-consistency checks. Functional testing verifies the main user actions (upload, parameter changes, ledger filters). Calculation verification compares values displayed by the dashboard with values re-derived from them – for example, checking that the regression equation reproduces the expected values listed in the outlier panel. Integration testing confirms that tiles, charts, rule lists and the ledger agree with each other.

6.2 Test Cases

The status column distinguishes cases that were **observed** in the deployed application and its screenshots from cases that must still be **verified** by the project team before final submission.

Table 13: Test cases

ID	Scenario	Input / action	Expected result	Status
TC-01	Load application	Open the project URL	Landing screen with upload bar, “Load sample”, sample.csv and new_batch.csv appears	Pass (observed)
TC-02	Load dataset	Load smart_grocery_warehouse_dataset.csv	Summary shows 3,133 SKUs across 15 categories and the four headline tiles	Pass (Fig. 5.1)
TC-03	Performance chart	Open Overview	Five technique bars on a 0–100 scale with matching metric cards	Pass (Fig. 5.1)
TC-04	Clustering view	Select X = Quantity_In_Stock, Y = Monthly_Sales, k = 4	Four coloured clusters with sizes summing to 3,133	Pass (Fig. 5.2)
TC-05	Association rules	Min support 0.10, min confidence 0.60	296 rules with support, confidence and lift	Pass (Fig. 5.2)
TC-06	Regression and outliers	$\|z\| > 2.5$	Equation and R^2 shown; 52 products flagged	Pass (Fig. 5.2)
TC-07	Ledger filters	Click ALL / NEW / OUTLIERS	Counts 3,133 / 0 / 52 are shown	Pass (Fig. 5.3)
TC-08	Parameter change	Move k, support, confidence and z sliders	Only the related panel and ledger update, consistently	To be verified
TC-09	Add new products	Upload new_batch.csv	New rows get class, cluster and outlier status; NEW count increases	To be verified
TC-10	Invalid file	Upload a file with fewer than two numeric columns	A clear message explains the requirement	To be verified
TC-11	Export	Click EXPORT in the ledger	Ledger is downloaded	To be verified
TC-12	Responsive view	Open on mobile width	Layout remains usable	To be verified

6.3 Evaluation Metrics

Each technique is judged with a metric suited to its task:

- **Classification:** accuracy (share of correct predictions) and F1-score (harmonic mean of precision and recall), evaluated with leave-one-out validation.
- **Clustering:** silhouette coefficient, from −1 (wrong cluster) through 0 (overlapping clusters) to +1 (well separated).

- **Association rules:** number of rules, average confidence, and lift for each rule.
- **Regression:** coefficient of determination R^2, the share of variance in Monthly_Sales explained by Quantity_In_Stock.
- **Outlier detection:** number and percentage of products flagged at the chosen |z| threshold.

Data Accuracy (%) = (Values correctly displayed / Values checked) × 100

6.4 Results and Discussion

6.4.1 Overall scorecard

Table 14 and Figure 6.1 summarise the headline results reported by the dashboard. Because the techniques use different metrics, the dashboard rescales them to 0–100 for display only; the bars should not be read as a ranking of which algorithm is "best".

Table 14: Headline results on the 3,133-product dataset

Technique	Raw metric	Normalised score	Interpretation
kNN classification	Accuracy 16.8%, F1 14.7%	16.8	Low, but about 2.5× random guessing
k-means (k = 4)	Silhouette 0.08	54.0	Weak cluster separation
Apriori	296 rules, avg confidence 81.7%	81.7	Strong rules, but many are trivial
Linear regression	$R^2 = 0.52$	51.9	Moderate explanatory power
Outlier detection	52 of 3,133 flagged	1.7	Small, focused exception list

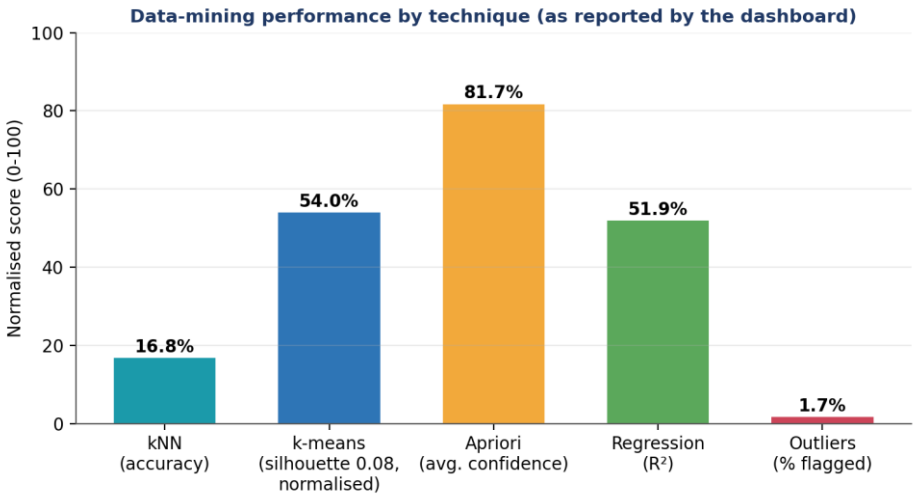


Figure 6.1: Data-mining performance by technique (values from the dashboard)

6.4.2 Classification with kNN

The kNN classifier predicts the product category from numeric attributes with leave-one-out accuracy of **16.8%** and F1 of **14.7%** over **15 classes**. If the 15 categories were equally likely, random guessing would be right about 1/15 = **6.7%** of the time, so the classifier performs about **2.5 times better than chance** (Figure 6.2). Nevertheless, it is wrong for five out of six products.

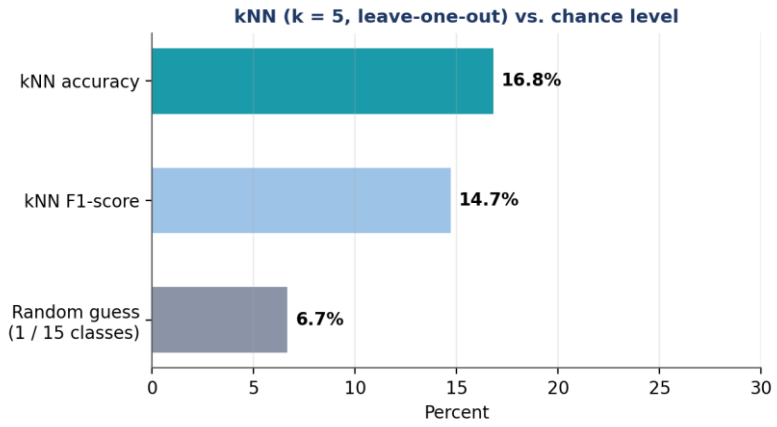


Figure 6.2: kNN accuracy and F1-score compared with random guessing

The product ledger (Figure 5.3) illustrates the errors. In the 12 rows visible in the screenshot only 3 products were predicted correctly (FreshMart Tomato, DailyChoice Sunflower Oil and ValueCart Moong). The others were confused with unrelated categories, for example Milk predicted as Bakery, Toilet Cleaner as Pulses and Sponge as Beverages. Prediction confidence ranged from 20% to 61%, meaning that the five neighbours rarely agreed.

Table 15: Sample of the product ledger (12 rows visible in Figure 5.3)

Product	Actual	Predicted	Conf.	Cluster	Stock	Monthly sales	Result
FreshMart Tomato	Vegetables	Vegetables	20%	C4	154	91	Correct
ValueCart Urad Dal	Pulses	Beverages	46%	C4	184	127	Wrong
DailyChoice Sunflower Oil	Cooking Oil	Cooking Oil	42%	C3	81	113	Correct
ValueCart Chilli	Spices	Bakery	40%	C4	104	137	Wrong
ValueCart Moong	Pulses	Pulses	42%	C3	109	114	Correct
ValueCart Olive Oil	Cooking Oil	Personal Care	36%	C2	90	105	Wrong
PrimePick Milk	Dairy	Bakery	39%	C1	109	63	Wrong
NatureNest Cookies	Bakery	Snacks	58%	C4	146	108	Wrong
FarmFresh Instant Mix	Packaged Foods	Beverages	43%	C1	65	97	Wrong
FreshMart Energy Drink	Beverages	Packaged Foods	21%	C1	18	28	Wrong
ValueCart Sponge	Household	Beverages	41%	C1	73	119	Wrong
FreshMart Toilet Cleaner	Cleaning Products	Pulses	61%	C3	253	168	Wrong

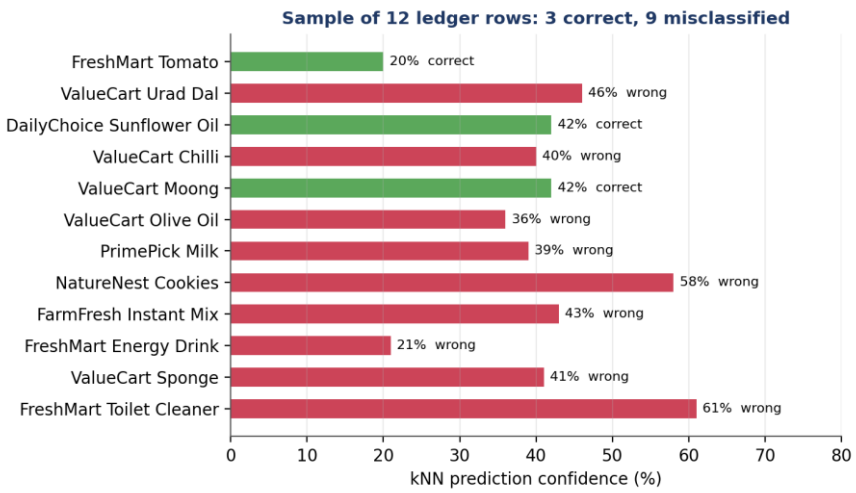


Figure 6.3: Prediction confidence for the 12 sample ledger rows (green = correct, red = wrong)

Why is accuracy low? Categories such as Pulses, Spices and Cooking Oil are defined by what a product is, while the 21 attributes describe prices, stock, sales and logistics. A cheap pulse and a cheap biscuit can have similar numbers. The 12-row sample (25% correct) is only a visual sample and should not be generalised; the overall figure remains 16.8%.

6.4.3 Clustering with k-means

With k = 4 the products split into clusters of 1,013, 585, 442 and 1,093 products (32.3%, 18.7%, 14.1% and 34.9%; Figure 6.4). The dashboard's legend names the clusters after a category (C1 Beverages, C2 Cooking Oil, C3 Cooking Oil, C4 Beverages), presumably the most common category inside each cluster.

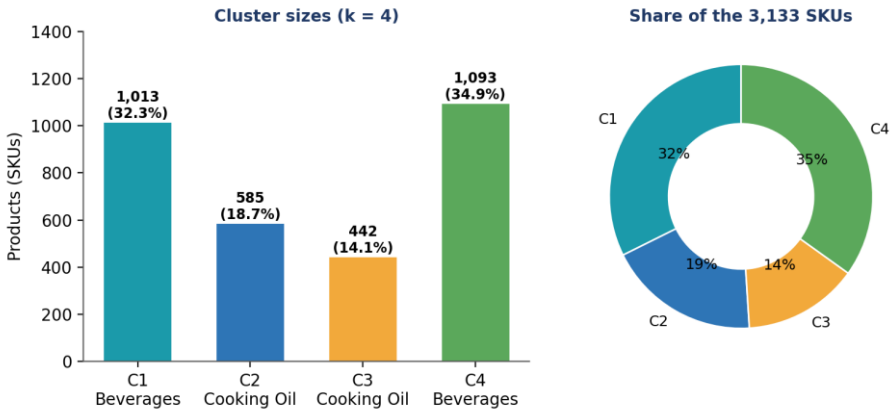


Figure 6.4: Cluster sizes and share of products (k = 4)

Two observations follow. First, two clusters are labelled Beverages and two Cooking Oil, so the clusters do **not** correspond to product categories: they group products by **stock-and-sales behaviour** (for example high stock and high sales versus low stock and low sales). Second, the silhouette coefficient is only **0.08**. On the usual interpretation scale (Figure 6.5) a value below 0.25 means that no substantial cluster structure has been found – the points form one continuous, wedge-shaped cloud (Figure 5.2) that k-means cuts into slices rather than separating natural groups. The

normalised score of 54.0 on the performance chart is simply (0.08 + 1)/2 × 100 and should not be mistaken for "54% good".

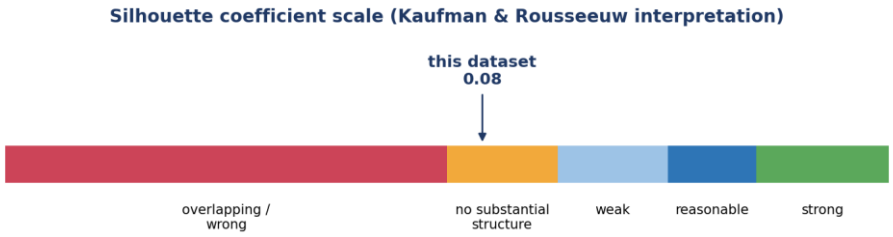


Figure 6.5: Position of the dataset's silhouette value (0.08) on the interpretation scale

6.4.4 Association rules with Apriori

At minimum support 0.10 and minimum confidence 0.60 the algorithm reports **296 rules** with an average confidence of **81.7%**. The strongest rules visible in the dashboard are listed in Table 16.

Table 16: Association rules shown in the dashboard

Rule (antecedent → consequent)	Support	Confidence	Lift
Monthly_Sales = Mid → Sales_Velocity = Mid	0.33	1.00	3.03
Sales_Velocity = Mid → Monthly_Sales = Mid	0.33	1.00	3.03
Monthly_Sales = High → Sales_Velocity = High	0.33	1.00	3.03
Sales_Velocity = High → Monthly_Sales = High	0.33	1.00	3.03
Demand_Level = High → Monthly_Sales = High	0.11	1.00	3.03
Demand_Level = High → Sales_Velocity = High	0.11	1.00	3.03

Interpretation. A confidence of 1.00 means that whenever the antecedent is true, the consequent is always true. The lift of 3.03 equals 1/0.33, which shows that each Low / Mid / High level covers about one third of the products – consistent with tertile-based discretisation. Because Sales_Velocity and Demand_Level are derived from sales, these rules mostly restate that the same quantity has been expressed in two ways. They are statistically strong but **not new knowledge**. The 296-rule total is therefore inflated by redundant rules, and the 81.7% average confidence overstates how much has been discovered.

6.4.5 Linear regression

Regressing Monthly_Sales on Quantity_In_Stock gives:

$$Monthly_Sales = 0.515 \times Quantity_In_Stock + 45.70 \qquad R^2 = 0.52$$

- **Slope 0.515:** each additional unit in stock is associated with about 0.5 additional units of monthly sales – products with more stock tend to sell more.
- **Intercept 45.70:** the expected monthly sales of a product with no stock; it is an extrapolation rather than a real-world value.
- **R^2 = 0.52:** stock level explains about 52% of the variation in monthly sales; the correlation coefficient is r = $\sqrt{0.52} \approx 0.72$, a strong positive relationship.

- **Residual spread:** the z-scores and residuals in the outlier list imply a residual standard deviation of about **31 units** (for example, (258 − 100.29) / 5.1 ≈ 30.9). This is derived from the displayed values, not read from the application.

Correlation between stock and sales does not prove that stock drives sales: popular products are probably stocked in larger quantities because they sell, so causation may run the other way.

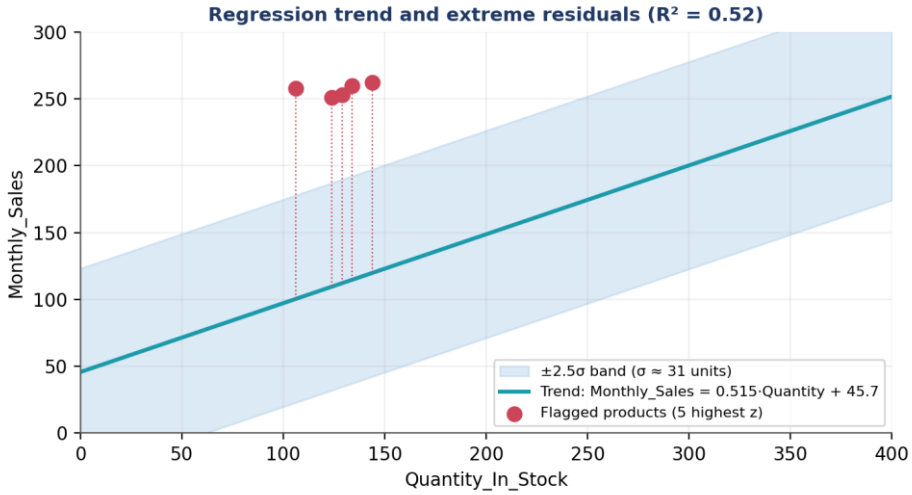


Figure 6.6: Regression trend with the ±2.5σ band and the five most extreme flagged products

6.4.6 Outlier detection

Using a threshold of |z| > 2.5, the dashboard flags **52 of 3,133 products (1.7%)**. Each of the five most extreme outliers sells roughly 140–158 units per month more than its stock level predicts (Table 17). The stock quantity implied by each trend value is obtained by inverting the regression equation, (trend − 45.70) / 0.515, and gives whole numbers (106, 134, 144, 124 and 129), which confirms that the equation and the listed values are consistent.

Table 17: The five highest-scoring outliers

Product	Actual monthly sales	Expected (trend)	Residual	z-score	Implied stock
DailyChoice Floor Cleaner	258.00	100.29	+157.71	5.1	106
ValueCart Turmeric	260.00	114.71	+145.29	4.7	134
PrimePick Turmeric	262.00	119.86	+142.14	4.6	144
PrimePick Cookies	251.00	109.56	+141.44	4.6	124
DailyChoice Tea	253.00	112.13	+140.87	4.5	129

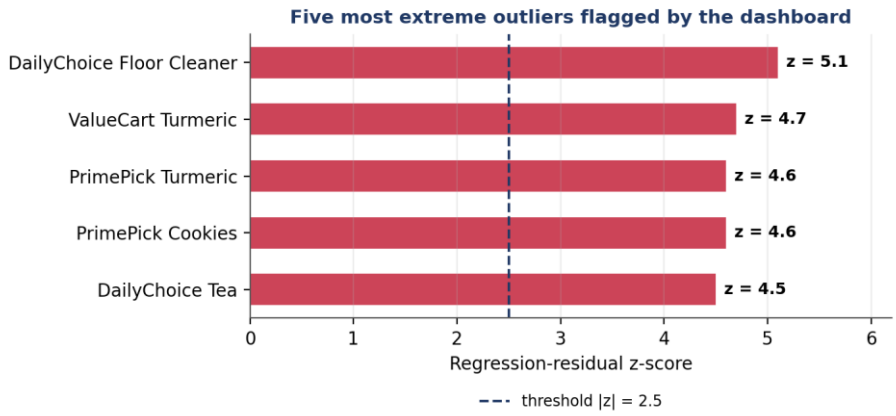


Figure 6.7: z-scores of the five most extreme outliers against the 2.5 threshold

The flagged products are all **above** the trend: they sell far more than their stock suggests, so they are potential **stock-out risks** that may need earlier or larger reorders. A manager can act on this list directly.

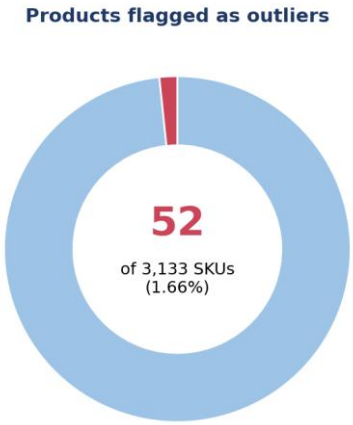


Figure 6.8: Share of products flagged as outliers

Is 1.7% reasonable? If residuals were perfectly normal, a two-sided threshold of $|z| > 2.5$ would flag 1.24% of products (about 39 of 3,133), and only 0.62% (about 19) would lie above the trend. The observed 1.66% (52) and the one-sided pattern of the top-ranked cases show that residuals have a heavier upper tail than a normal distribution, as is typical for sales data where a few products sell disproportionately well (Figure 6.9).

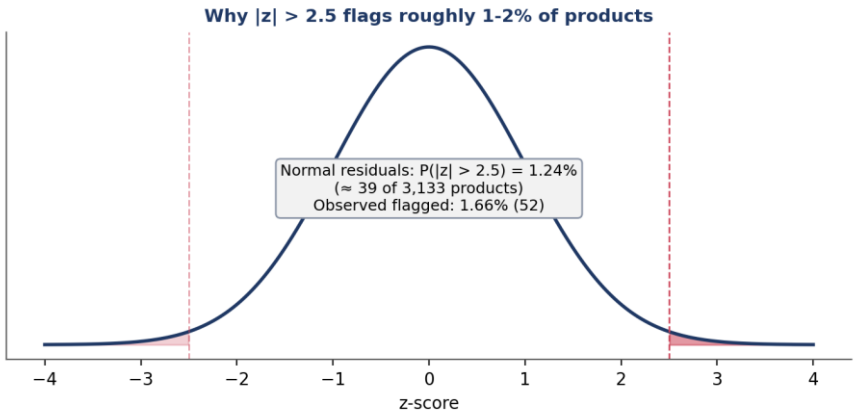


Figure 6.9: Expected tail probability for |z| > 2.5 compared with the observed flag rate

6.5 Key Findings

1. The dashboard turns a 3,133-row, 21-attribute product file into five complementary views in a single page, and keeps tiles, charts and the ledger consistent.
2. **kNN** reaches 16.8% accuracy (F1 14.7%) on 15 classes – about 2.5 times chance – showing that numeric price, stock and sales attributes carry only limited information about the product category.
3. **k-means** produces clusters (silhouette 0.08) that reflect stock-and-sales behaviour rather than category, and the data forms one continuous cloud rather than distinct groups.
4. **Apriori** finds 296 high-confidence rules, but the strongest ones (confidence 1.00, lift 3.03) relate derived attributes to each other and carry little new insight.
5. **Regression** shows a strong positive relationship between stock and monthly sales (R^2 = 0.52, $r \approx 0.72$) with a residual spread of about 31 units.
6. **Outlier detection** isolates 52 products (1.7%) that sell far above their stock-based expectation, giving the manager a short, actionable exception list.

6.6 Recommendations for Improvement

Table 18: Observed limitation and suggested improvement

Observation	Suggested improvement	Expected benefit
Low kNN accuracy (16.8%)	Scale or standardise attributes; select the most informative features; try decision tree or random forest; use k-fold cross-validation and report a confusion matrix	Higher accuracy and a clearer view of which categories are confused
Weak clusters (silhouette 0.08)	Standardise features, test several k with the elbow and silhouette methods, cluster on 3–4 selected attributes	More meaningful product segments
Trivial association rules	Exclude derived attributes (Sales_Velocity, Demand_Level) and rank by lift; add transaction-basket data	Genuinely new, actionable rules
Single-predictor regression	Use multiple regression with price, discount, rating and purchase frequency; add residual plots	Higher R^2 and better outlier thresholds
Fixed |z| threshold	Use robust statistics (median and MAD) and show both above- and below-trend cases	Fewer false alarms

Observation	Suggested improvement	Expected benefit
Static sample data	Connect to live point-of-sale and stock data with scheduled refresh	Operational, up-to-date decisions

6.7 Result Evaluation Areas

Table 19: Result evaluation areas

Evaluation area	Verification approach	Expected contribution
Dashboard loading	Open the deployed application	Confirms availability
Metric accuracy	Compare tiles, cards and ledger counts (3,133 / 52)	Confirms consistency of calculations
Regression consistency	Invert the equation for each listed outlier	Confirms the displayed equation and residuals
Parameter sensitivity	Vary k, support, confidence and z threshold	Shows how stable the results are
Ranking usefulness	Inspect the outlier list with a manager	Supports managerial action
Responsive design	Test varied screen widths	Supports accessibility

6.8 Learning Outcomes

- Applying five mining techniques to one business dataset and deploying the result as a web prototype.
- Understanding why a result's metric must be interpreted against a baseline (e.g. 16.8% against 6.7% chance).
- Recognising that high-confidence rules can be trivial when attributes are derived from each other.

7. CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The Data Deli Manager Performance Dashboard (Meridian WMS) was developed as a web-based business-intelligence and data-mining prototype for deli and grocery performance monitoring. It shows how a raw product dataset of 3,133 SKUs can be transformed into a dataset summary, product categories, behavioural clusters, association rules, a sales trend and an exception list within a single interactive page.

The case study demonstrates core data-warehousing and data-mining principles: structured analytical data, dimensional modelling, five mining techniques and metric-based evaluation. The critical analysis shows that tool output must be read carefully: kNN accuracy is modest although well above chance, k-means finds weak structure, the strongest rules are partly redundant, and regression and outlier detection give the most directly actionable results for a manager.

The deployed application is available at https://data-deli-manager-performance.vercel.app/. Before operational use, the system should be connected to verified business data sources and strengthened with authentication, data security, role-based access, audit trails and validated reporting rules.

7.2 Limitations

- The prototype uses a prepared sample dataset rather than a live data feed.
- Classification accuracy is low, and the kNN evaluation uses leave-one-out validation without a confusion matrix.
- Clusters have weak separation (silhouette 0.08) and are sensitive to the chosen k and axes.
- Association rules include trivial rules from derived attributes.
- Regression uses one predictor and assumes a linear, normally distributed residual pattern.
- No integration with payment, payroll, accounting or live inventory systems.
- Access control and data privacy need additional work for production deployment.
- KPIs must be aligned with actual business policies and objectives.

7.3 Future Scope

- Integrate real-time point-of-sale, inventory and supplier data with automated refresh and alerts.
- Add decision-tree, random-forest and cross-validated classification with confusion matrices.
- Improve clustering with feature scaling, automatic selection of k and per-cluster profiles.
- Mine true market-basket rules from transaction data to support cross-selling and bundles.
- Extend regression to multiple predictors and add demand forecasting and reorder recommendations.

- Add waste, spoilage and shelf-life monitoring using the expiry and storage attributes.
- Provide report export, scheduled manager summaries and mobile alerts.
- Add role-based access, audit logging and secure authentication.
- Support multiple outlets for regional or franchise-level comparison.

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